POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor and Computer Architecture

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between Microprocessor and Microcontroller with examples. 7

   b) What are the different types of FLAGS used in 8085 microprocessor? Show the status for the Flags after performing the following operation 5F H+ 02 H. 8

2. a) What is machine cycle? Explain the timing diagram of an instruction LXI H, 3050H. 7

   b) Write a 8085 ALP to find the largest number from an array of given 10 numbers stored at location starting from COOOH and store the result in the memory location 2024H. 8

OR

Write a 8085 ALP to find smallest of three number.

3. a) Write an ALP to find the smallest number in a data array. Data from location C000H to C005H. 7

OR

Explain ALP development tools in 8086 microprocessor.

   b) What There is a very simple CPU for the given set of Instructions: 8

Instructions	Instruction Code	Operations
OR	00AAAAAA	AC←AC∨M[AAAAAA]
AND	01ĀAAAAA	AC←AC ∧ M[AAAAAA]
CMP	10AAAAAA	AC←AC'
LDA	11XXXXXX	AC← M[AAAAAA]

Let the instruction width be 8-bits and address is 6 bits. Design the CPU's Register Section, State Diagram and ALU.

4. a) Design the hardwired control unit for the CPU described in question 3 b. 7

b) Design a micro-sequencer control unit with horizontal microcode for the given CPU in question number 3b. 8

5. a) Trace the multiplication of 4 and -7 using RTL code of Booth's algorithm. Also verify the obtained product. 8

b) Explain the Set- Associative technique during the mapping of Cache memory and compare it with associative mapping. 7

6. a) How DMA reduces the CPU engagement dealing I/O interrupts with DMA than programmed I/O and interrupt driven I/O. Explain. 7

b) Describe Flynn's Taxonomy. Explain the different topologies used in multiprocessor communication. 8

7. Write short notes on: **(Any two)** 2×5

a) RISC Vs. CISC

b) Hardwired Vs. micro programmed CU

c) Computer Organization Vs. Computer Architecture